MA1522 Linear Algebra for Computing

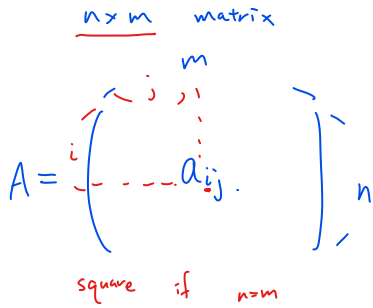
Lesson 3: Matrices, Matrix Operations and Matrix Equations

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson3` (e.g. `A1234567Z_lesson3.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 3**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev





Basic rules:

- Please (mainly) ask the questions related to the course.

Section 2.1 Definition and Special Types of Matrices

An $m \times n$ **matrix** is a rectangular array $A = (a_{ij})_{m \times n}$ of m rows and n columns; a_{ij} is the entry in row i, column j — row index first, always. Its **size** is $m \times n$.

column vector

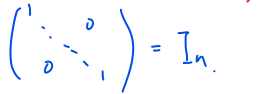


Two matrices are **equal** when they have the same size and the same entries in every position. An $n \times 1$ matrix is a **column vector** and a $1 \times n$ matrix a **row vector**; unqualified, "vector" means column vector.

row vector

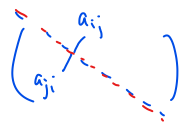
$(\quad\quad\quad)$

If $m = n$ the matrix is **square**, of **order** n. Some square shapes:



- **diagonal**: $a_{ij} = 0$ whenever $i \neq j$; the **identity** I_n is the diagonal matrix with every $a_{ii} = 1$;
- **upper triangular**: $a_{ij} = 0$ whenever $i > j$; **lower triangular**: $a_{ij} = 0$ whenever $i < j$; **strictly** so if the diagonal is zero too;
- **symmetric**: $a_{ij} = a_{ji}$ for all i, j.





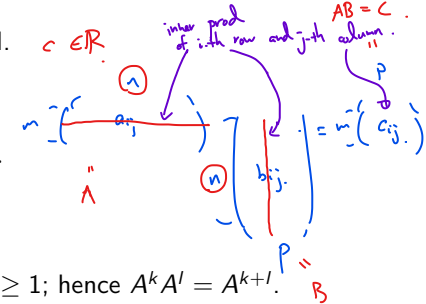
sym. w.r.t. anti-diagonal



Section 2.2 Matrix Algebra: The Operations

- **Addition**: $A + B = (a_{ij} + b_{ij})_{m\times n}$ — same size required.
- **Scalar multiplication**: $cA = (c\,a_{ij})_{m\times n}$ — always defined.
- **Multiplication**: for A of size $m \times n$ and B of size $n \times p$,

$$AB = \left(\sum_{k=1}^{n} a_{ik} b_{kj}\right)_{m\times p}.$$



- **Transpose**: $A^T = (a_{ji})_{n\times m}$ — reflect in the diagonal.
- **Power**: for A square ($n \times n$) only, $A^0 = I_n$ and $A^k = AA^{k-1}$ for $k \geq 1$; hence $A^k A^l = A^{k+l}$.

$$\underbrace{m \times n}_{A} \qquad \underbrace{n \times p}_{B} \quad \Longrightarrow \quad AB \text{ has size } m \times p$$

Question: Why this definition of multiplication is "correct" or "interesting"?

Section 2.2 What Carries Over from Numbers, and What Does Not

Still true. For matrices of compatible sizes and $c \in \mathbb{R}$:

- $A + B = B + A$, $(A + B) + C = A + (B + C)$;
- $(AB)C = A(BC)$;
- $A(B + C) = AB + AC$, $(A + B)C = AC + BC$;
- $c(AB) = (cA)B = A(cB)$;
- $AI_n = A = I_m A$, $A + 0 = A$, $A0 = 0$;
- $(A + B)^T = A^T + B^T$, $(cA)^T = cA^T$;
- $(AB)^T = B^T A^T$. $(ABC)^T = C^T B^T A^T$

No longer true. In general:

- $AB \neq BA$ — one may be defined and the other not;
- $(AB)^T \neq A^T B^T$ — the order reverses;
- $AB = 0$ does not force $A = 0$ or $B = 0$;
- hence $AB = AC$ does not force $B = C$.

Every cancellation habit you have from $\mathbb{R}$ needs re-examining.

$0 = AB - AC = A(B - C)$

Example 1

$A = (a_{ij})_{n\times n}$

Let A and B be square matrices of order n. Which of the following are always true?

true 1. $(A + I_n)^2 = A^2 + 2A + I_n$.

false. 2. $(A + B)^2 = A^2 + 2AB + B^2$.

$(A+I_n)^2 = (A+I_n)(A+I_n)$

$= A\cdot A + I_n\cdot A + A\cdot I_n + I_n\cdot I_n$

$= A^2 + A + A + I_n = A^2 + 2A + I_n$.

$(A+B)(A+B)$

$= A^2 + A\cdot B + B\cdot A + B^2$

Compare both sides, the equality holds iff $AB = BA$.

Question 1

Let $A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$.

$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = BA$ $\quad$ $AB = A = BA$ $\quad$ $AB = A^2 = BA$

$B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ $\quad$ $s \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\quad$ $t \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$

I_2

1. Without computing anything, write down two matrices B with $AB = BA$.

2. Find all matrices B (2×2) with $AB = BA$.

$AB = BA$

$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x & y \\ z & w \end{pmatrix} = \begin{pmatrix} x & y \\ z & w \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$

$\begin{pmatrix} z & w \\ 2x+3z & 2y+3w \end{pmatrix} = \begin{pmatrix} 2y & x+3y \\ 2w & z+3w \end{pmatrix}$

Question 1 — solution

$$\begin{cases} z = 2y \\ w = x+3y \\ 2x+3z = 2w \\ 2y+3w = z+3w \end{cases} \xrightarrow{\text{std form}} \begin{cases} -2y+z = 0 \\ -x-3y+w = 0 \\ 2x+3z-2w = 0 \\ 2y-z = 0 \end{cases} \Rightarrow \left(\begin{array}{cccc|c} 0 & -2 & 1 & 0 & 0 \\ -1 & -3 & 0 & 1 & 0 \\ 2 & 0 & 3 & -2 & 0 \\ 0 & 2 & -1 & 0 & 0 \end{array}\right)$$

$\Downarrow$ ref.

$$\left(\begin{array}{cccc|c} 1 & 0 & \frac{3}{2} & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & 0 \\ 6 & 6 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 \end{array}\right)$$

pivot columns

$z = t$. $w = s$.

$x = -\frac{3}{2}t + s$. $\quad y = \frac{1}{2}t$.

$$B = \begin{pmatrix} -\frac{3}{2}t+s & \frac{1}{2}t \\ t & s \end{pmatrix} = s\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{t}{2}\begin{pmatrix} -3 & 1 \\ 2 & 0 \end{pmatrix}$$

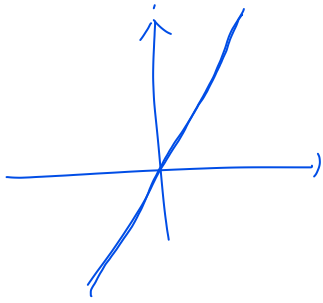
$$= \underbrace{\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}}_{A} + \begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix}.$$

Question 1 — remark

$\mathbb{R}^2 \subseteq \mathbb{R}^4$

$\{$ analogue

$\mathbb{R}^1 \subseteq \mathbb{R}^2$



$\mathbb{R}^2$.

Section 2.3 Linear Systems and Matrix Equations

$$\star: \begin{cases} 2x + 2y + 3z = 1 \\ x + 4y + 5z = 2 \end{cases}$$

A linear system in standard form is exactly the single **matrix equation**

$$A\mathbf{x} = \mathbf{b}, \qquad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix},$$

$$A = \begin{pmatrix} 2 & 2 & 3 \\ 1 & 4 & 5 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\star = A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = b.$$

with **coefficient matrix** $A = (a_{ij})_{m \times n}$, **variable vector** $\mathbf{x}$ and **constant vector** $\mathbf{b}$.

Writing $A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix}$ by columns — $\mathbf{a}_i$ is the **coefficient vector** of x_i — this is the **vector equation**

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}.$$

Question 2

Consider the linear system

$$\begin{cases} x_1 + 2x_2 + x_3 + 3x_4 = 8 \\ x_1 - 2x_2 - x_3 + x_4 = -2 \\ 2x_1 - x_2 - x_3 + 7x_4 = 1 \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & -2 & -1 & 1 \\ 2 & -1 & -1 & 7 \end{pmatrix} \quad b = \begin{pmatrix} 8 \\ -2 \\ 1 \end{pmatrix}$$

1. Write it as a matrix equation $A\mathbf{x} = \mathbf{b}$ and as a vector equation.
2. Solve it, and write the general solution in the form $\mathbf{u} + t\mathbf{v}$.
3. Write $A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{pmatrix}$ by columns. Hence find real numbers c_1, c_2, c_3 with

$$\mathbf{a}_4 = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + c_3\mathbf{a}_3.$$

Question 2 — solution: parts (1) and (2)

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 8 \\ 1 & -2 & -1 & 1 & -2 \\ 2 & -1 & -1 & 7 & 1 \end{array}\right) \xrightarrow{\text{ref}} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & 4 & 0 \\ 0 & 0 & 1 & -7 & 5 \end{array}\right)$$

A $\quad$ b. $\quad$ pivot columns

$x_1 = 3 - 2t, \; x_2 = -4t, \; x_3 = 5 + 7t.$ $\quad$ $x_4 = t \qquad t \in \mathbb{R}.$

$$\begin{pmatrix} 3 - 2t \\ -4t \\ 5 + 7t \\ t \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 0 \\ 5 \\ 0 \end{pmatrix}}_{u} + t \underbrace{\begin{pmatrix} -2 \\ -4 \\ 7 \\ 1 \end{pmatrix}}_{v}$$

u: particular sol.

$A \cdot (u + tv) = b \quad (*)$

$\begin{cases} Au = b & \text{(take } t = 0\text{)} \\ Atv = 0. \end{cases}$

$(*) \Rightarrow Au + Atv = b.$

$\Rightarrow Atv = 0. \quad Av = 0.$

Question 2 — solution: part (3)

Section 2.3 Block Multiplication

The same idea, one column at a time. If $B = \begin{pmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{pmatrix}$ has columns $\mathbf{b}_j$ and A has rows $\mathbf{r}_1, \ldots, \mathbf{r}_m$, then

$$AB = A \begin{pmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{pmatrix} = \begin{pmatrix} A\mathbf{b}_1 & \cdots & A\mathbf{b}_p \end{pmatrix}, \qquad AB = \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \end{pmatrix} B = \begin{pmatrix} \mathbf{r}_1 B \\ \vdots \\ \mathbf{r}_m B \end{pmatrix}.$$

j-th column of $AB = A \times (\, j$-th column of $B)$; $\quad i$-th row of $AB = (\, i$-th row of $A) \times B$.

Question 3

Let $A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$. $\begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \\ x_{41} & x_{42} & x_{43} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ solve for x_{ij}.

Find a 4×3 matrix X with $AX = I_3$. Is it unique?

$$A \begin{pmatrix} x_{11} \\ x_{21} \\ x_{31} \\ x_{41} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

$$A \begin{pmatrix} x_{12} \\ x_{22} \\ x_{32} \\ x_{42} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$A \begin{pmatrix} \vdots \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Question 3 — solution

$$\left(\begin{array}{cccc|c|ccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{\text{ref}} \left(\begin{array}{cccc|c|ccc} 1 & 0 & 0 & 2 & 1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

A

$$X = \begin{pmatrix} 1-2t_1 & -1-2t_2 & 1-2t_3 \\ t_1 & 1+t_2 & -1+t_3 \\ -t_1 & -t_2 & 1-t_3 \\ t_1 & t_2 & t_3 \end{pmatrix} \iff AX = I_3$$

$t_1, t_2, t_3 \in \mathbb{R}$.

Challenge

Question 3 produced infinitely many X with $AX = I_3$. Explain why, for the same A, no 4×3 matrix Y satisfies $YA = I_4$.

Next lesson

- **Sections 2.4–2.6 Inverses, Elementary Matrices and Their Uses**

Please watch the video clips for these sections beforehand.

If you do not know how to upload your submission, come to ask me after class.